

MOHAMMAD TAAHA ASHRAF

Full-Stack Developer - Backend Systems & AI Tooling

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TECHNICAL SKILLS

Languages:TypeScript, JavaScript, C++, SQL, Python

Frontend:Next.js 15, React, Tailwind CSS, Base UI, HTML5, CSS3

Backend & Realtime:Node.js, Express, FastAPI, Socket.io, WebRTC, WebSockets

Databases & Caching:PostgreSQL, MongoDB, Redis, MySQL

Tools & DevOps:Turborepo, pnpm, Docker, Nginx, Git, GitHub, GitLab, Postman, Vercel, Render, Joi, Jest, Supertest

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor | CropChain (Aperture 2.0) | [GitHub Repository](#)

Tech Stack: Node.js, Express, Joi, Jest, Supertest, JavaScript

- Transformed an unprotected backend route into a secure, production-ready API by designing Joi validation schemas and a centralized validateRequest middleware to intercept invalid payloads.
- Engineered a robust automated integration test suite using Jest and Supertest to simulate live API endpoints and validate error-handling path coverage.

PROJECTS

Musafir Trips | Full-Stack Travel Agency Platform | [Live Demo](#) | [GitHub](#) | *Tech Stack: Next.js, TypeScript, Prisma, PostgreSQL (Neon), Auth.js (v5), Tailwind CSS*

- Built a full-stack CMS-driven travel platform with a secure admin dashboard covering tours, blogs, hero slides, enquiries, awards, and site-wide settings — deployed on Vercel.
- Implemented NextAuth v5 (JWT) with Google OAuth and credential-based login, protected by lightweight Edge middleware for route-level auth.
- Used Next.js Server Actions with Prisma 7 for all database mutations, with automatic ISR revalidation (revalidatePath) on every content change.
- Integrated Cloudinary for image uploads across all admin forms and a Tiptap rich-text editor for blog and content management.

exac.draw | Collaborative Whiteboard | [Live Demo](#) | [GitHub](#) | *Tech Stack: Next.js 15, Turborepo, WebSockets, PostgreSQL, Tailwind CSS, pnpm*

- Designed a high-performance monorepo** using Turborepo to manage shared packages between frontend and backend, reducing local build times and improving cross-environment code sharing.
- Engineered **real-time canvas synchronization** using **WebSockets**, enabling multi-user simultaneous collaboration with minimal latency.
- Implemented a custom canvas rendering engine to handle complex geometric shapes alongside persistent state architecture in PostgreSQL.

Nitpick | AI-Powered PR Review GitHub App | [GitHub Repository](#) | *Tech Stack: Node.js, TypeScript, BullMQ, Redis, Octokit, Gemini API*

- Building a GitHub App that automatically reviews pull requests using the Gemini API, posting inline code-quality and logic feedback as PR comments.
- Architected an asynchronous job pipeline with BullMQ and Redis to queue and process PR reviews without blocking GitHub webhook response times.
- Integrated the GitHub REST API via Octokit to fetch diffs, post structured review comments, and manage installation-level permissions across repos.

kidsportal | Interactive Learning Platform | [GitLab](#) | *Tech Stack: FastAPI, Python, React, InkJS, MySQL*

- Collaborated in a team of three to design and build a comprehensive administrative dashboard for managing educational content, structural analytics, and user progress.

- Developed a custom "Stitcher" logic compiler to process database-stored story fragments into InkJS-compatible bytecode for dynamic, branching narratives.
- **Built a scalable backend using FastAPI** a scalable backend using FastAPI to manage real-time interactive story states and concurrent student user data efficiently.

EDUCATION

Jamia Hamdard University *New Delhi, India*

Bachelor of Technology in Computer Science and Engineering • Expected Graduation: 2027

Core Coursework: Data Structures and Algorithms (DSA) in C++, Object-Oriented Programming (OOP), Database Management Systems (DBMS), System Design basics.